

Project Initialization and Planning Phase

Date	15 March 2026
Team ID	
Project Title	EcoTrack – AI-Powered Carbon Footprint Tracker
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview

Objective

To develop an AI-powered web application that helps users monitor, analyze, and reduce their personal carbon footprint through intelligent habit tracking, carbon emission analysis, and personalized sustainability recommendations.

Scope

The project covers carbon footprint calculation for travel, food, and household energy usage. It integrates Firebase Authentication, Firebase Firestore, Carbon Interface API, Groq AI, and Chart.js to provide secure data storage, AI-powered recommendations, and interactive sustainability analytics.

Proposed Solution

Approach

Develop a full-stack web application using React.js, Flask, Firebase Firestore, and Groq AI to calculate carbon emissions, analyze user habits, provide AI-powered sustainability recommendations, and visualize environmental impact through interactive dashboards.

Key Features

- User Registration and Login using Firebase Authentication.
- Daily Habit Logger for Travel, Food, and Energy.

- Carbon Footprint Calculator using Carbon Interface API.
- AI-Powered Eco-Friendly Recommendations using Groq LLaMA 3.3-70B.
- Interactive Dashboard with Charts and Heatmaps.
- Personal Carbon Budget Tracking.
- Carbon Footprint History and Reports.
- Achievement Badges and Streak Tracking.
- Responsive User Interface with React.js and Tailwind CSS.

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware	Processor	Intel Core i5 (10th Gen) / AMD Ryzen 5 or above
	RAM	Minimum 8 GB (Recommended 16 GB)
	Storage	Minimum 256 GB SSD (Recommended 512 GB SSD)
	GPU (Optional)	NVIDIA GTX 1650 or above
Software	Frameworks	React.js, Flask
	Libraries	Firebase SDK, Axios, Chart.js, python-dotenv
	Development Environment	Visual Studio Code, Git, GitHub, Vite
Data	Database & APIs	Firebase Firestore, Carbon Interface API, Groq AI API
		(LLaMA 3.3-70B Versatile)